



Background of UAV Localization

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Outline of presentation

I. Background

II. Related Work

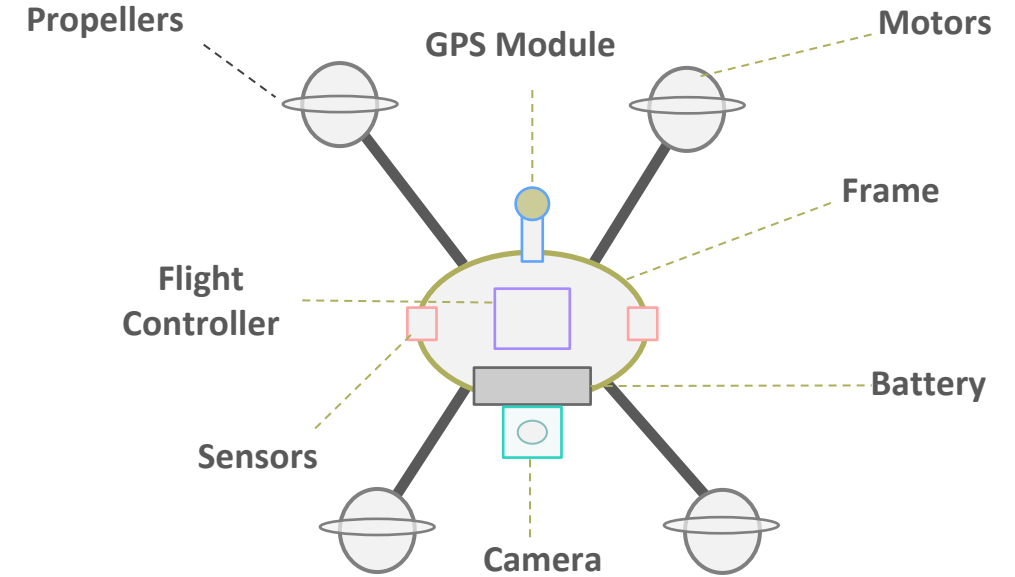
III. Device Free Indoor UAV Localization

IV. Conclusion

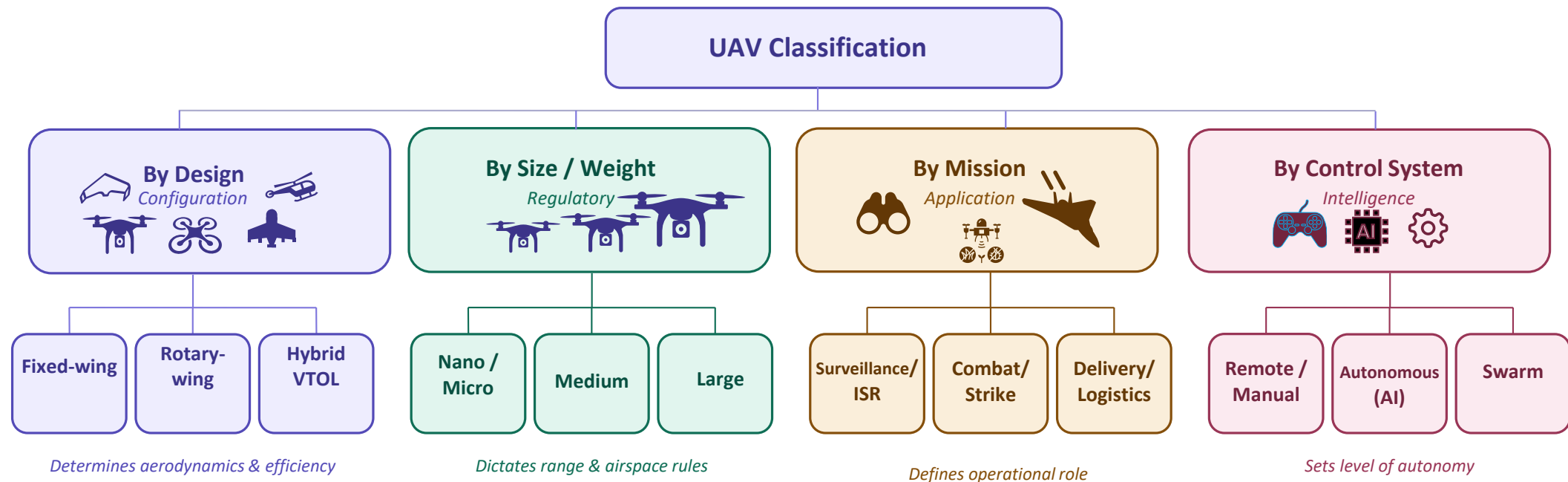
UAV (Unmanned Aerial Vehicle)

❑ A UAV is an aircraft that operates without an onboard human pilot, controlled either remotely or through autonomous systems.

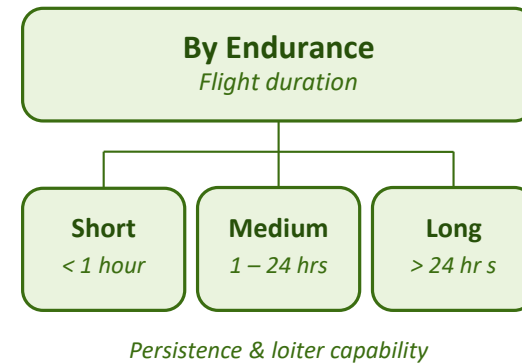
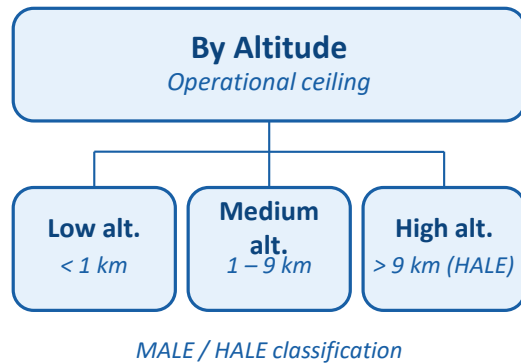
- Flight Controller (Brain of UAV)
- Sensors (IMU, Camera)
- Motors & Propellers (Movement)
- GPS Module (Navigation)



Classification of UAV

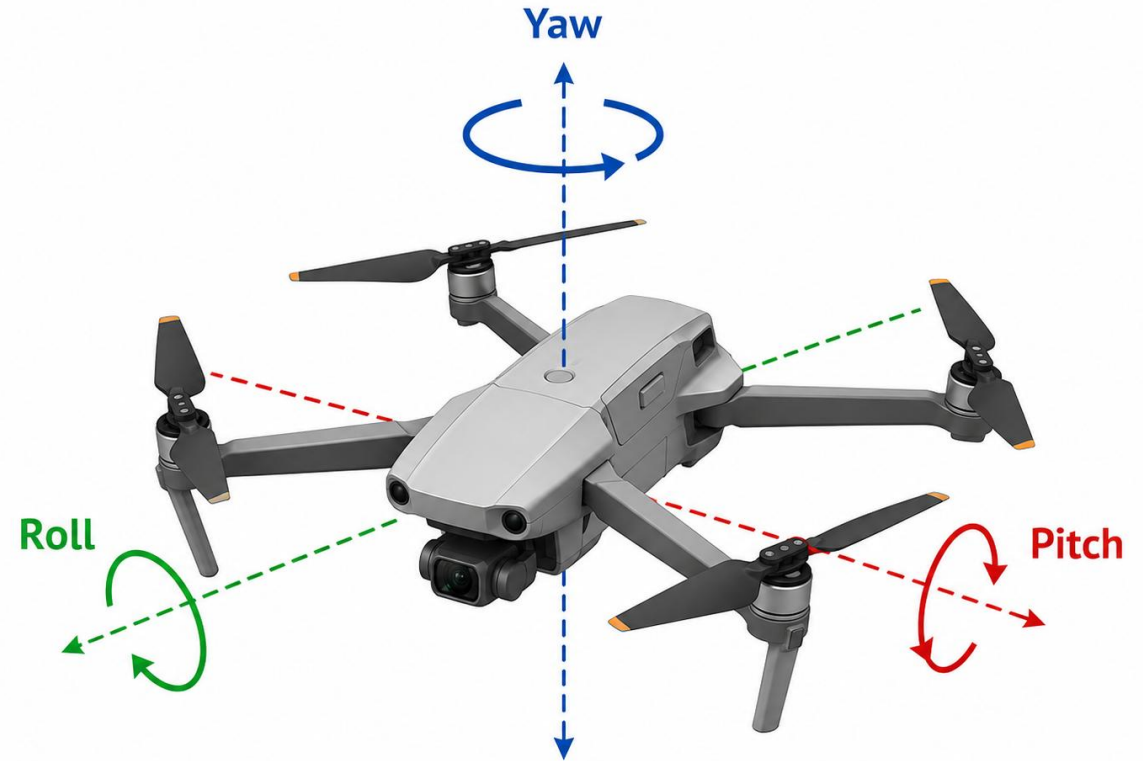


Additional classification dimensions



Localization of UAV

- ❑ UAV localization is the process of estimating the exact position and direction of a UAV during the flight.
- Estimate the UAV position (x,y,z) and orientation (roll, pitch , yaw)
- Necessary for : Autonomous Navigation, Path Planning, Obstacle Avoidance, Environment Mapping,, Target Tracking, Search & Rescue.
- UAV localization faces challenges such as signal loss , sensor errors, dynamic environments, obstacle avoidance , and maintaining accurate real time positioning.



Indoor vs Outdoor UAV Localization

UAV localization methods depend strongly on the environment.

➤ Outdoor UAV Localization :

- Outdoor UAV localization mainly relies on GPS/GNSS satellite signals.
- In outdoor environments, various sizes of UAVs can use to carry different types of sensors and payloads.



➤ Indoor UAV Localization :

- Indoor environments usually have weak GPS signals, Close obstacles, and dynamic obstacle.
- Indoor environment require the use of small UAV for safe and efficient navigations.
- Small UAV cannot easily carry heavy sensors such as LiDAR or RADAR.

Application of Indoor UAV Localization



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Indoor Positioning System



Inertial System

TECHNOLOGIES

- Accelerometer
- Gyroscope
- Magnetometer

TECHNIQUES

Strapdown

Body-frame integration

Magnetic Field Map

Fingerprint-based navigation



Network System

TECHNOLOGIES

- UHF-RFID
- Wi-Fi
- Ultra Wideband
- Bluetooth

TECHNIQUES

Signal Strength (RSS)

Received signal intensity

Signal Angle (AoA)

Angle of arrival estimation



Acoustic System

TECHNOLOGIES

- Audible Sound
- Ultrasound

TECHNIQUES

Time of Flight

Absolute propagation time

Echo (SONAR)

Reflection-based ranging



Vision System

TECHNOLOGIES

- Camera
- RADAR
- LiDAR

TECHNIQUES

Barcode Scanning

Marker-based identification

Visual odometry (VO)

SLAM

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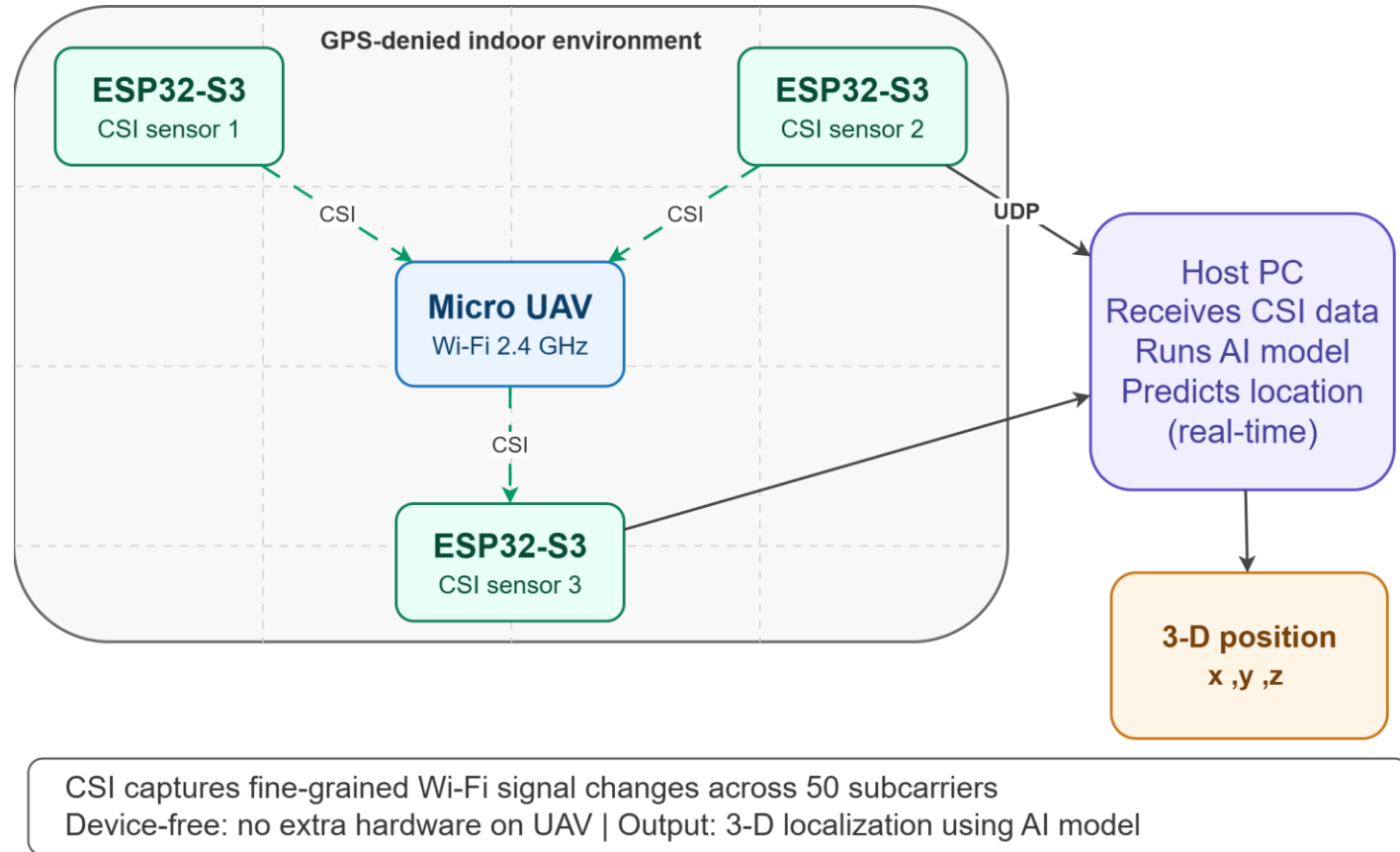
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Main Idea of the Paper

- Use UAV own wifi signal to estimate the localization of UAV .
- ❑ No extra hardware needed on the drone
- ❑ 3 low-cost ESP32-S3 sensors (≈\$7 each) are placed on the ceiling to capture the signal
- ❑ A deep learning model processes this signal and outputs the drone's x, y, z position
- ❑ Final Result of UAV 26 cm near to the actual position.



System Architecture

❑ **All devices are on the same Wi-Fi network. The drone doesn't know it's being tracked — the sensors just listen.**

1. UAV Wi-Fi Communication

- UAV uses Wi-Fi for normal flight control, video, and status data

2. CSI Data Collection

- 3 ESP32-S3 sensors capture CSI from UAV Wi-Fi packets

3. Data Transmission

- Each sensor sends extracted CSI data to the host PC via UDP

4. Signal Processing on Host PC

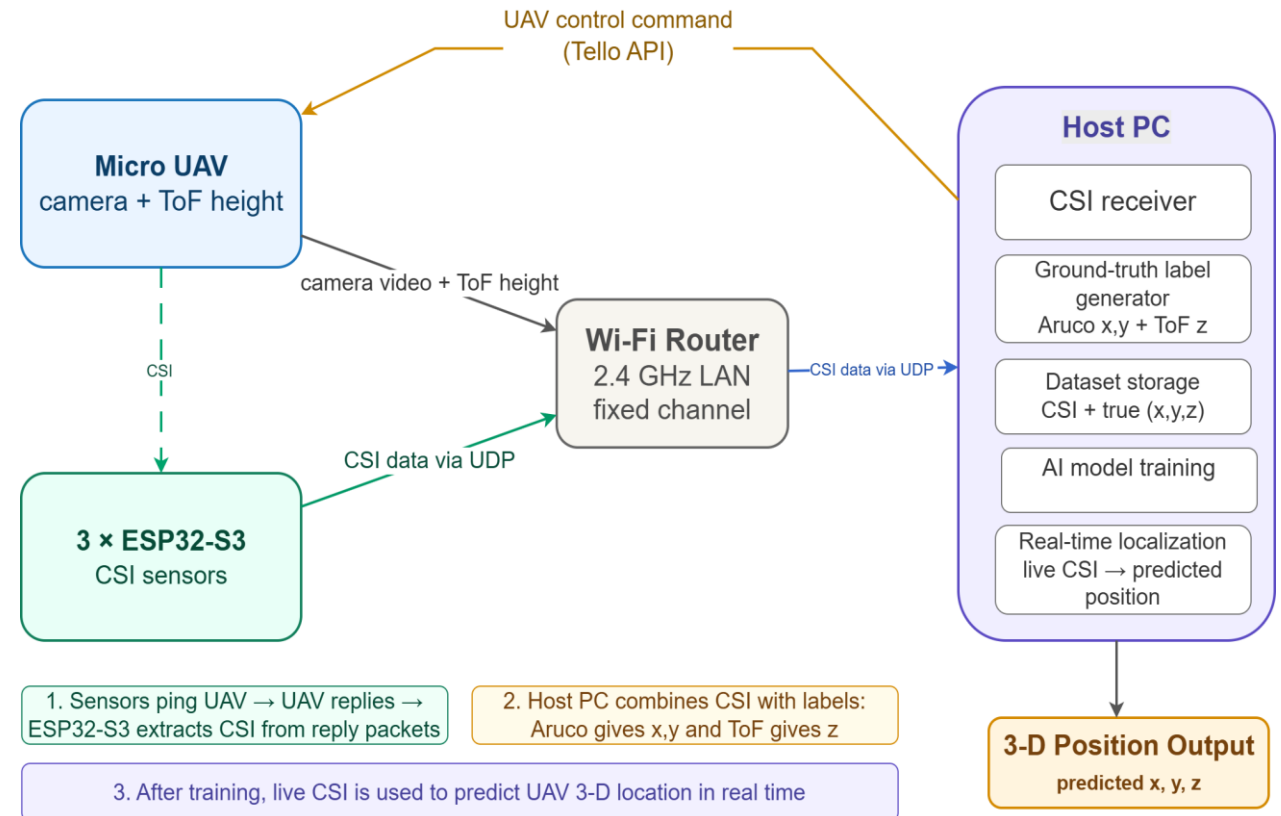
- RCSE cleans the raw CSI signal by reducing distortion and noise

5. AI-Based Localization

- SiS model predicts UAV 3-D position: (x, y, z)

6. Feedback Control

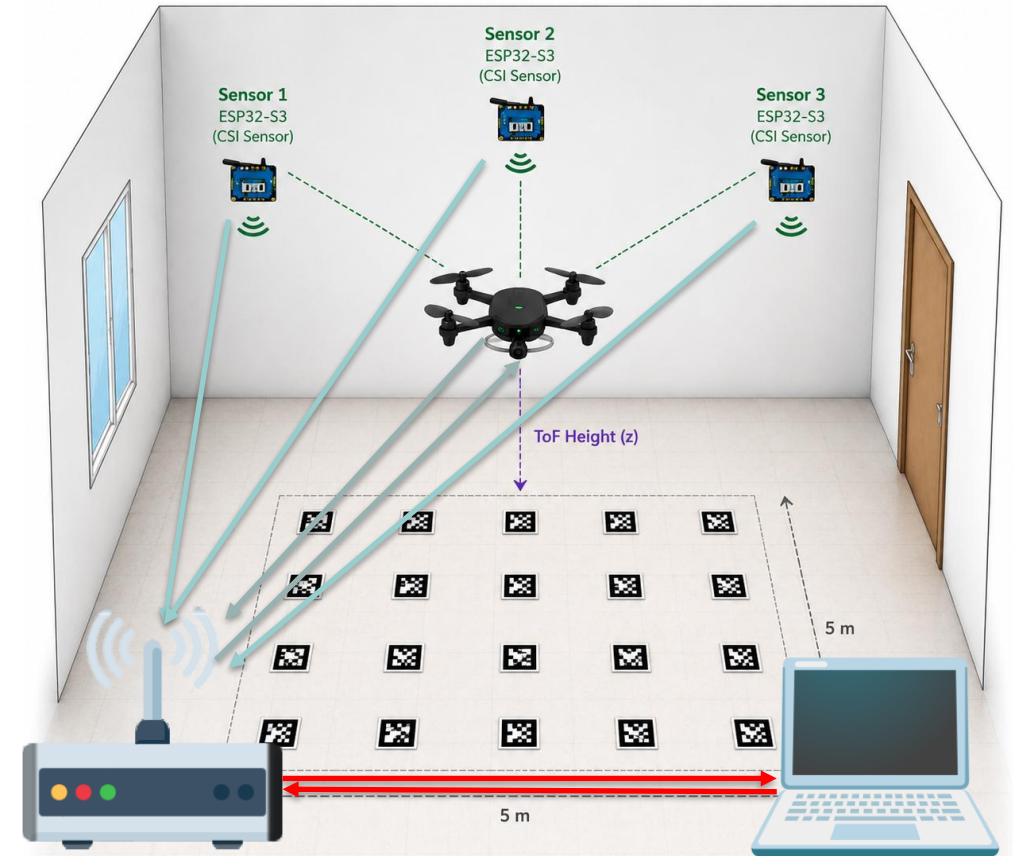
- Predicted position can be used by PID controller for closed-loop UAV guidance



Hardware Implementation Plan

- ❑ **The entire hardware setup costs around \$175. Nothing is attached to the drone — all sensing happens from the ceiling**

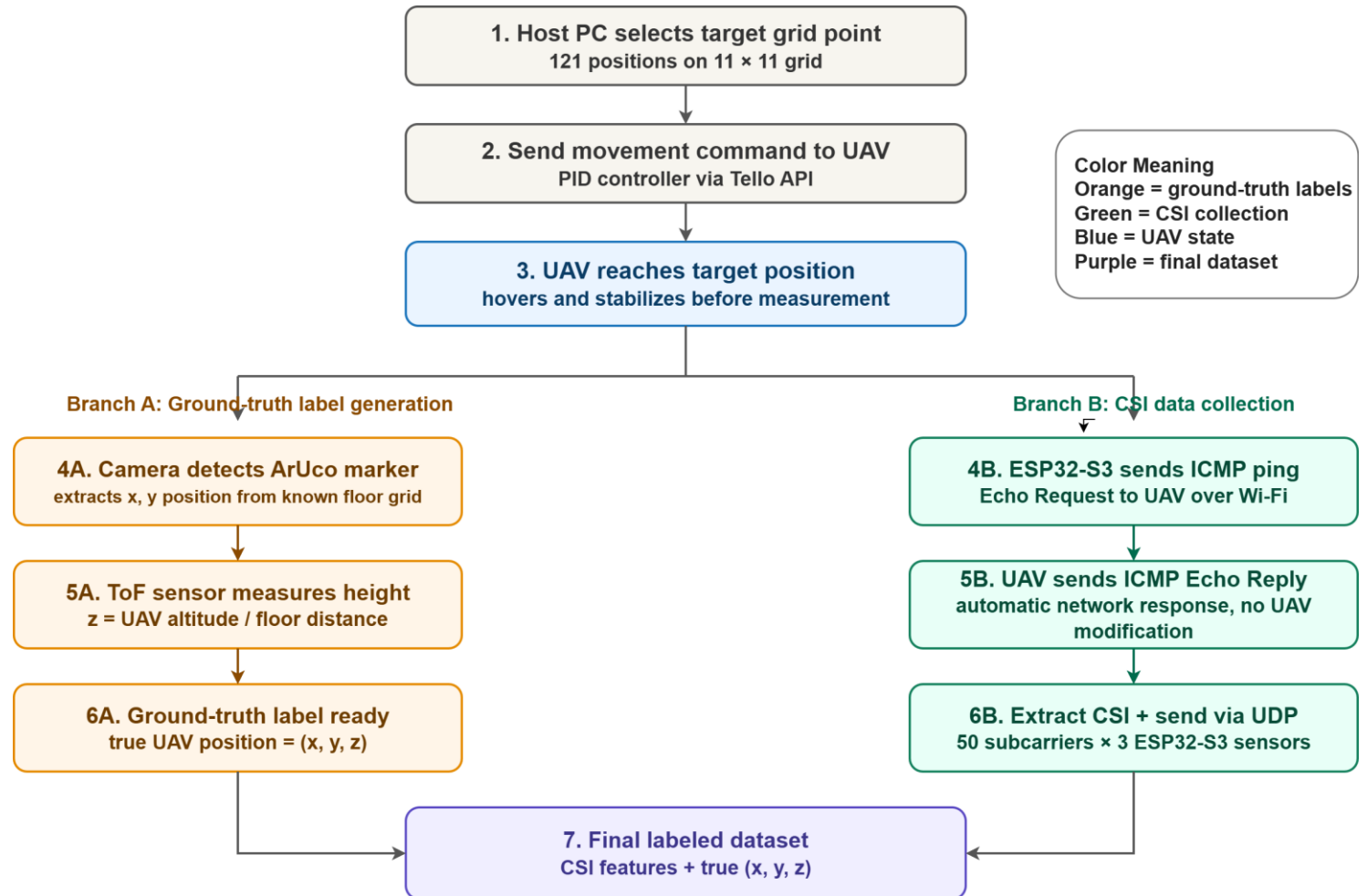
1. UAV Platform (DJI Tello UAV)
 - Wi-Fi-enabled drone controlled using Python SDK.
2. CSI Sensing Units (3 × ESP32-S3 CSI Sensors)
 - Ceiling-mounted in a triangular layout to capture UAV Wi-Fi CSI from different directions.
3. Network Infrastructure (Wi-Fi Router)
 - Connects the UAV, ESP32-S3 sensors, and host PC on the same 2.4 GHz network
4. Ground-Truth Labeling
 - ArUco Marker Grid: 11 × 11 printed floor markers used to obtain true x-y position.
 - ToF / Height Sensor Measures: UAV altitude and provides true z position
5. Processing Unit (Host PC)
 - Receives CSI data, runs RCSE signal cleaning, trains the SiS model, and performs real-time inference.



Data Collection workflow

- ❑ Data collection is fully automated. The drone visits each point, stabilizes, captures data, and moves on — no manual labeling needed

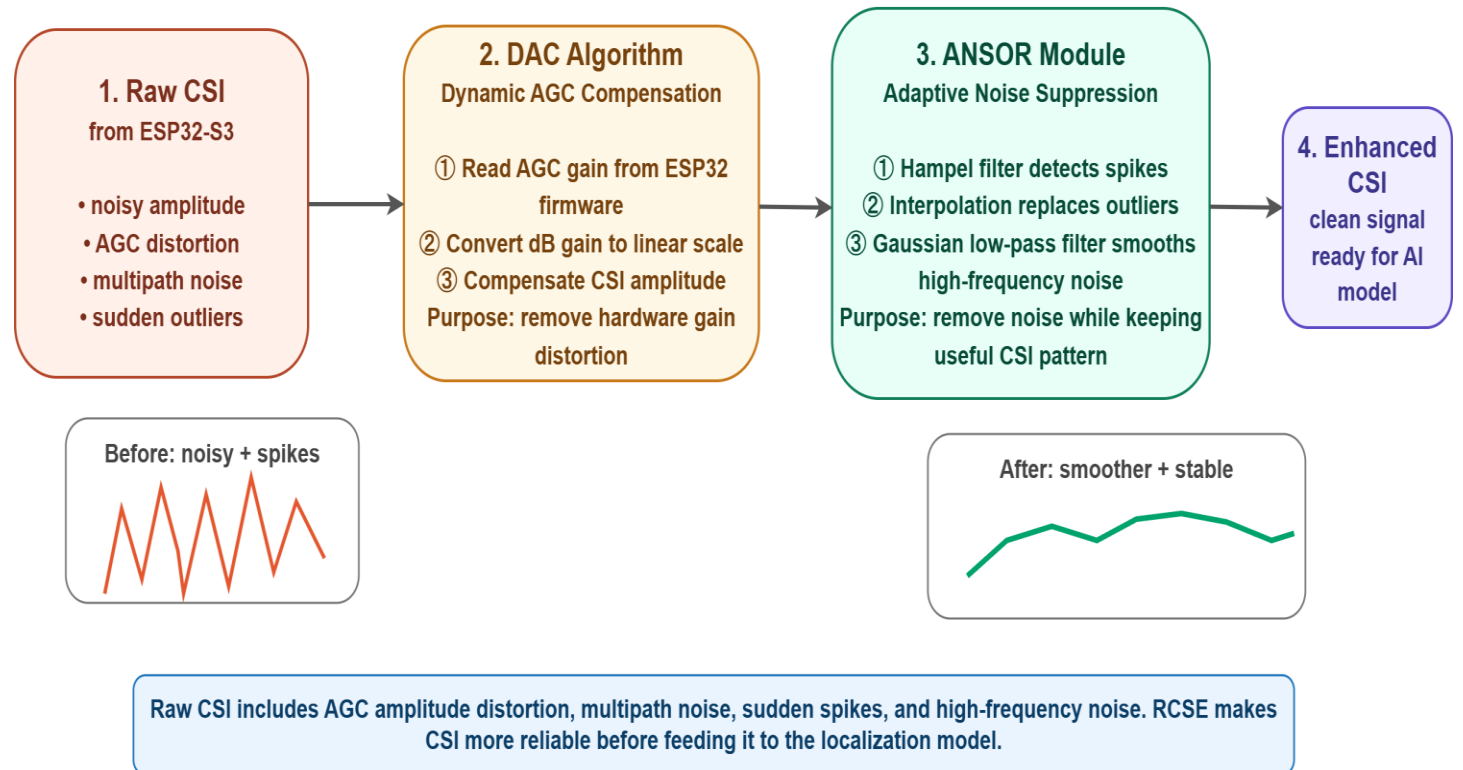
- UAV moves to each predefined grid point using PID control
- UAV stabilizes before data recording
- ESP32-S3 sensors send ICMP ping and collect CSI from reply packets
- UAV camera detects ArUco markers for x-y ground truth
- ToF sensor measures altitude for z ground truth
- Final dataset stores CSI features with true (x, y, z) labels



CSI Signal Cleaning (RCSE Framework)

- ❑ Raw CSI has two problems: AGC distortion and multipath noise
 - AGC changes signal gain automatically, which hides true CSI amplitude changes
 - DAC compensates AGC effect using the gain value from ESP32 firmware
 - Multipath creates reflections, spikes, and interference in CSI
 - ANSOR removes outliers using Hampel filtering and smooths noise using Gaussian filtering
- Final output: cleaner CSI for the SiS AI model

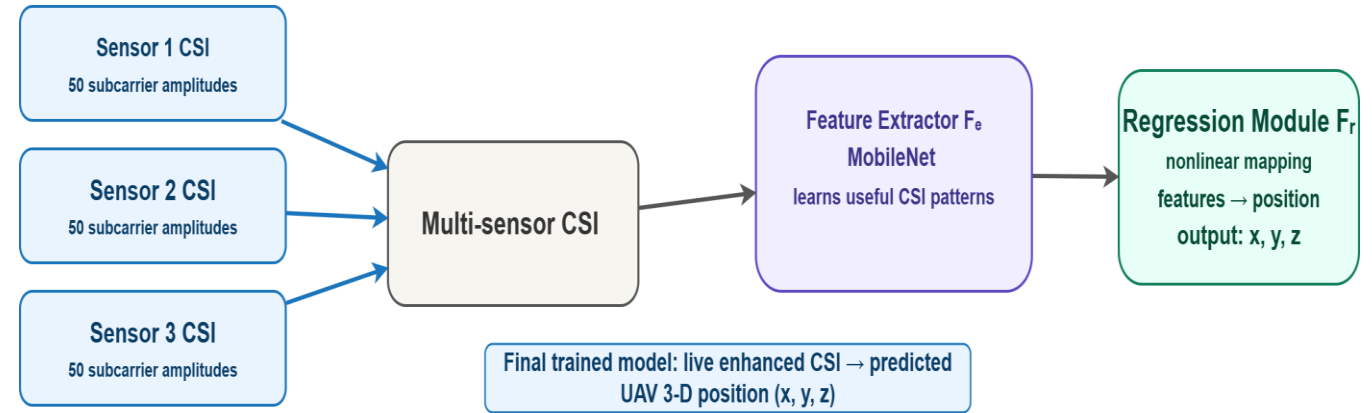
Raw CSI → Dynamic AGC Compensation → Noise and Outlier Removal → Enhanced CSI for AI model



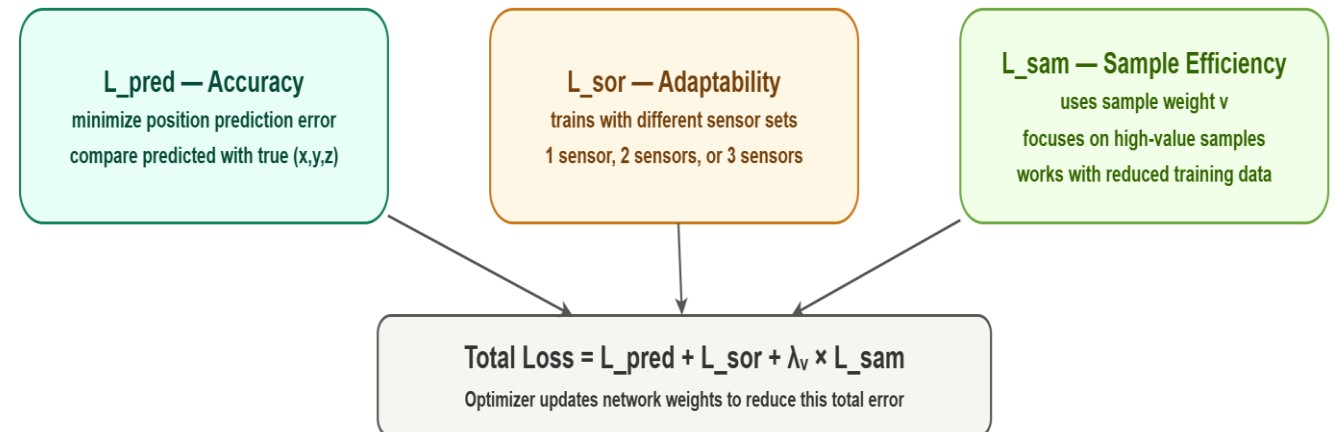
AI Model (SiS Architecture)

❑ The model takes clean CSI from 3 sensors as input and directly predicts the drone's x, y, z position. It's designed to work even if sensors are missing or data is limited.

- Input CSI: 3 sensors $\times$ 50 subcarrier amplitudes
- Feature extractor F_e learns useful CSI patterns from each sensor
- MobileNet 1D is used as a lightweight CSI feature extractor
- Features from 3 sensors are joined together by concatenation
- Regression head F_r predicts UAV position: (x, y, z)
- SiS improves robustness using different sensor combinations
- MOO loss trains the model for accuracy, sensor robustness, and data efficiency



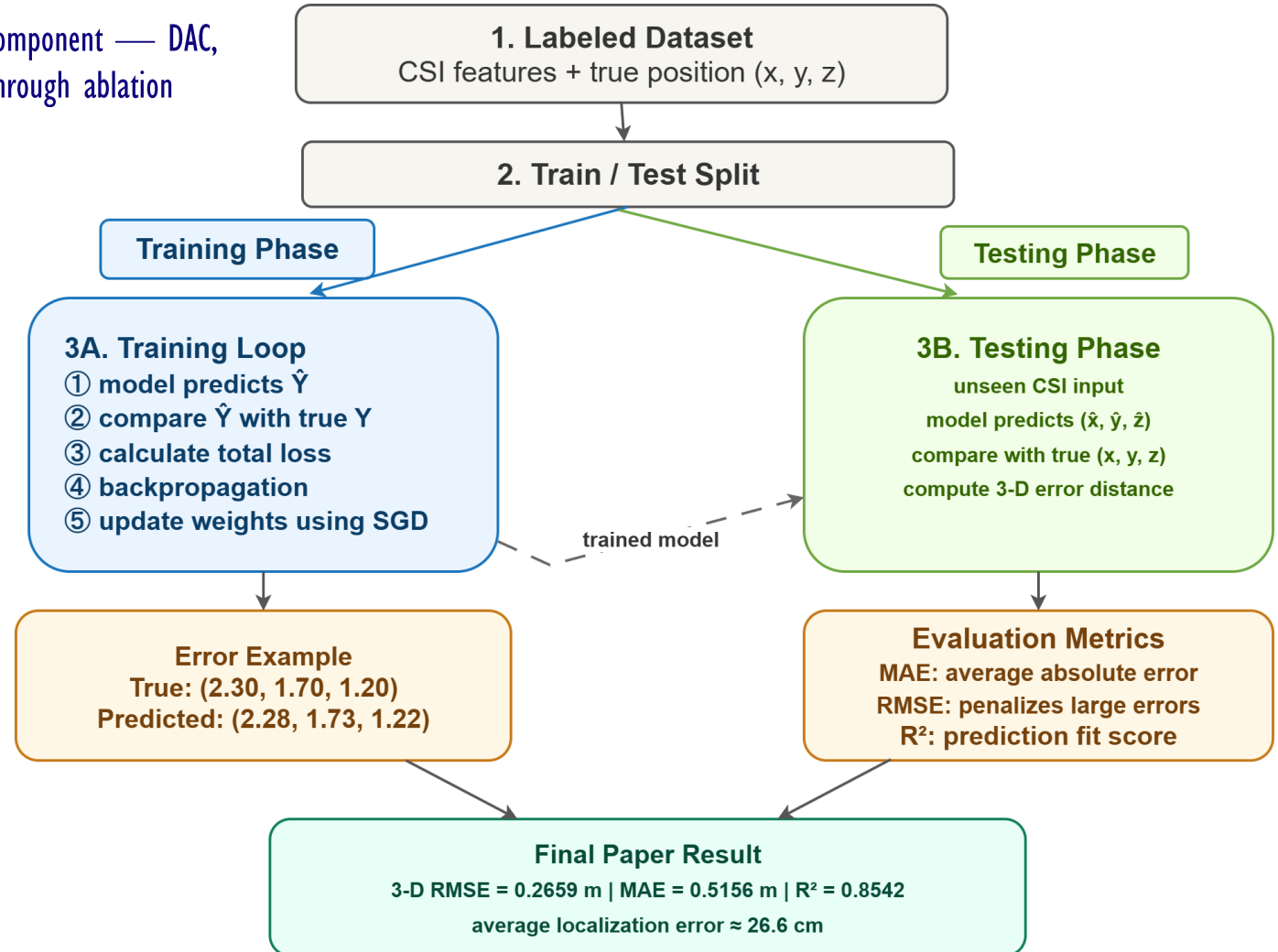
SiS: Sensor-in-Sample Multiobjective Optimization — three objectives trained jointly



Training and Evaluation

- ❑ The model is trained for 400 epochs on labeled CSI dataset. Every component — DAC, ANSOR, and SiS — adds measurable improvement, which confirmed through ablation

- Split labeled CSI data into train and test sets
- Train SiS model to learn $\text{CSI} \rightarrow (x, y, z)$
- Calculate loss between predicted and true position
- Update model weights using optimizer
- Test on unseen CSI samples
- Evaluate using MAE, RMSE, and R^2



Meaning: The model is trained with labeled CSI data. During testing, predicted UAV position is compared with true position. The final RMSE summarizes average 3-D localization error over all test samples.

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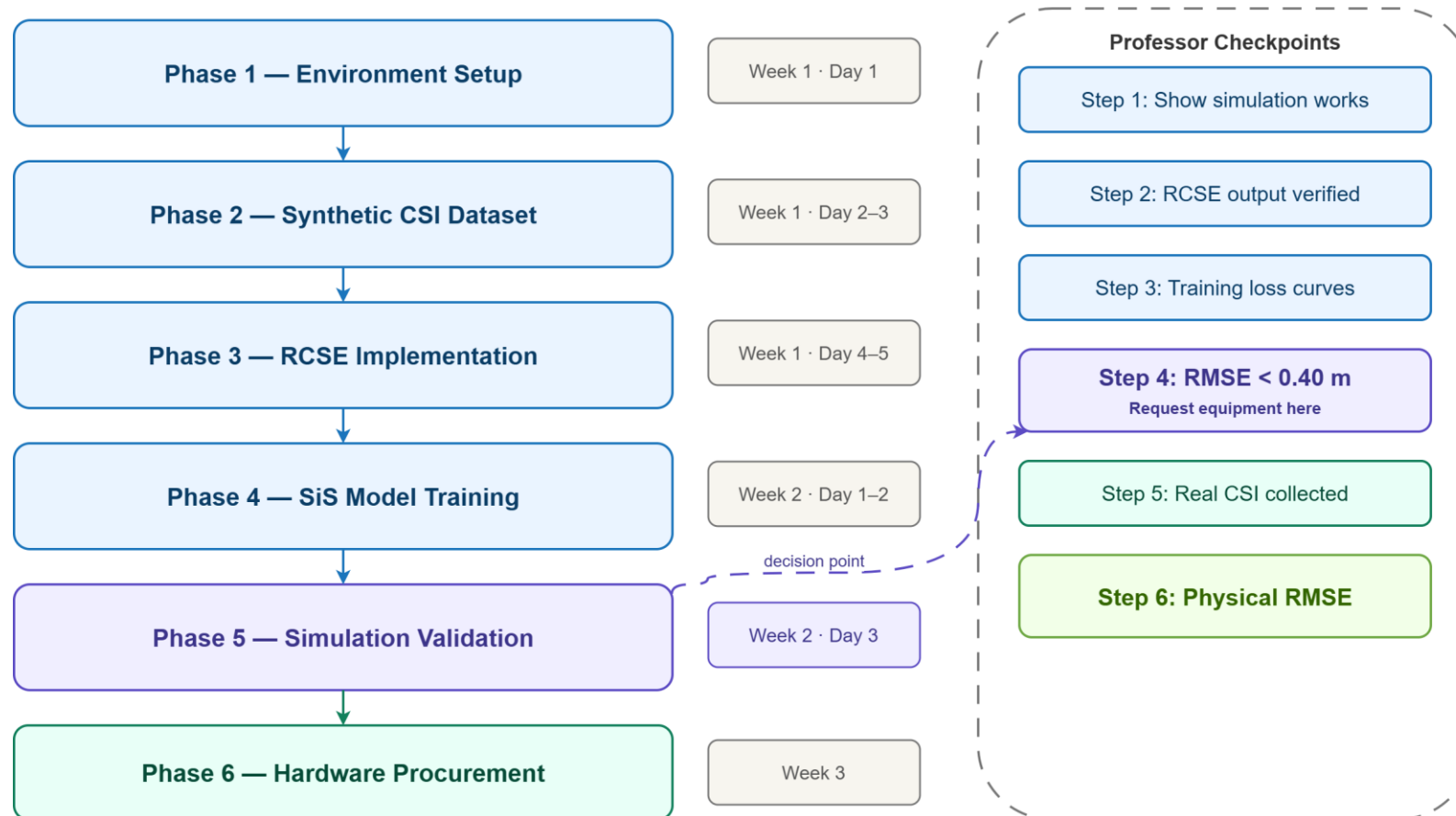
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Reproduction Roadmap

Experimental Reproduction Roadmap for CiUAV

Goal: prove implementation feasibility first in simulation, then request hardware for physical reproduction



Thank you for your listening!